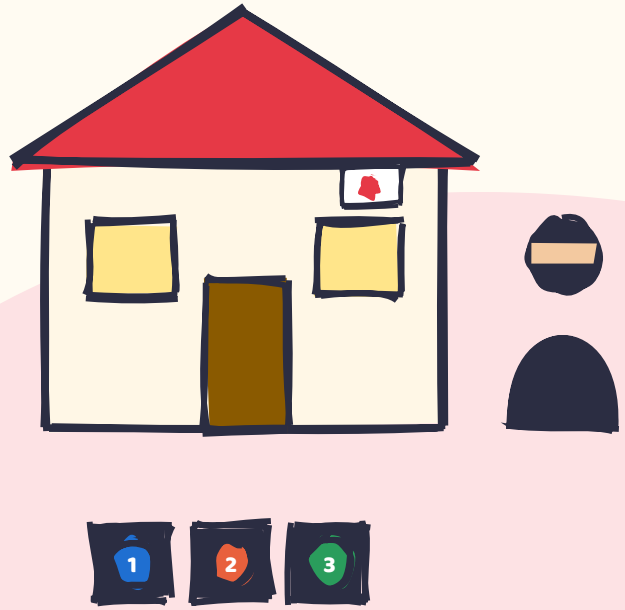
Junior Makers · Class 9

Home Burglar Alarm with a Secret Code

Only people who know the code can switch it off!

PIR motion sensing | 3-button code | Security logic



Did you know?
Most home alarms give you a few seconds after you open the door to type the code before the siren starts.

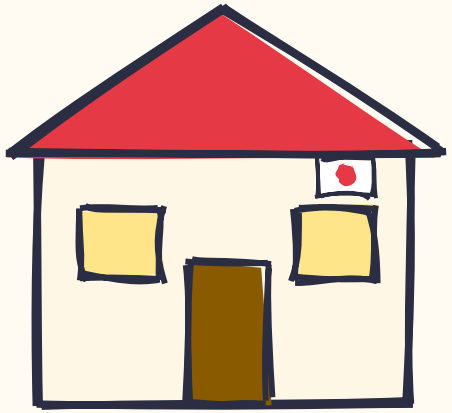
LEARN

Let's think...

Why does an alarm need a secret code?

A 2 am problem

The alarm is ON. Dad gets back late from work and opens the door.
WEE-OO WEE-OO! The whole street wakes up.
How can the alarm tell the **family** from a **burglar**?



who is it?

Recognise faces?

Use a key?

A secret code only family knows!

Today: we build an alarm that spots movement, and can only be switched off with a **3-button secret code**.

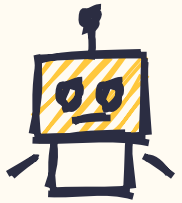
Learning aims

Today I will learn to...

1. Use a **PIR sensor** to detect people moving
2. Make the alarm **armed** or **disarmed**
3. Program a **3-button code** where the **order** matters
4. Use a **step counter** variable to remember progress
5. Choose and keep **codes and passwords** safe

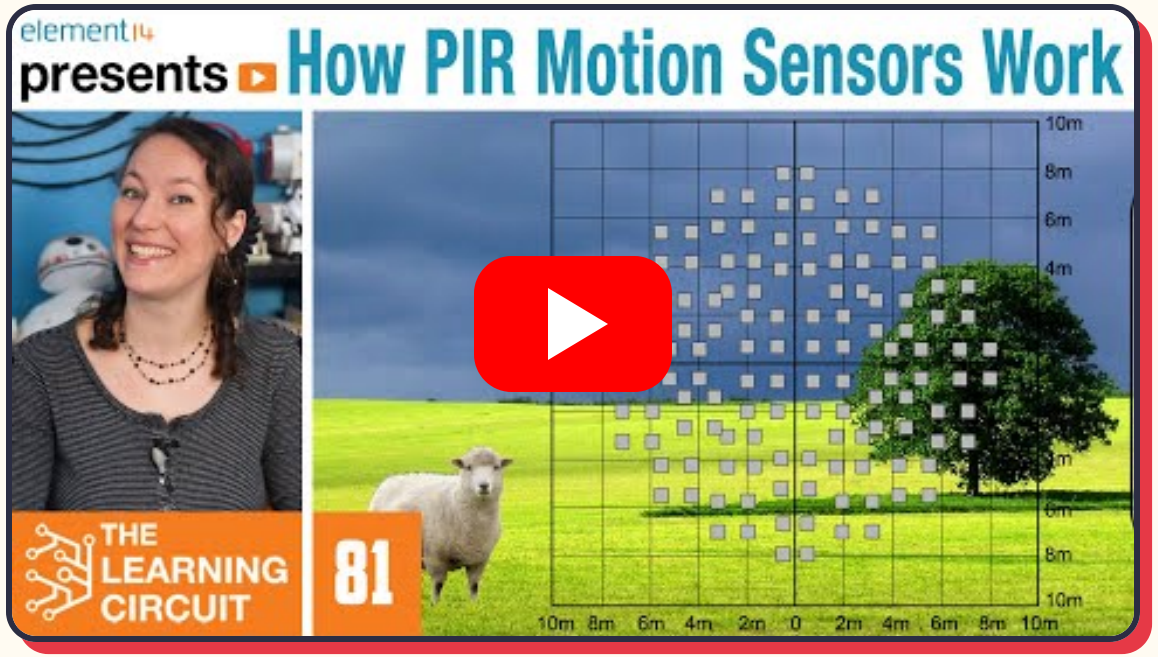
Keywords

PIR · motion · armed · disarmed · sequence · state · password



Watch and think

Video time!



How PIR Motion Sensors Work

element14 (The Learning Circuit) · about 7 minutes · youtu.be/jMnQmDgPCUg

Watch for...

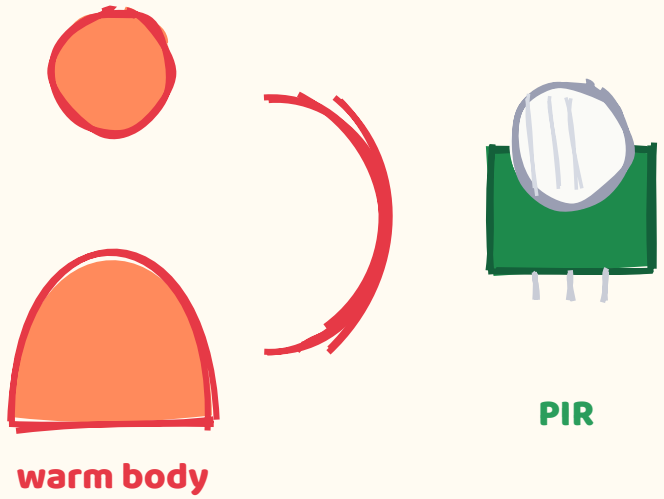
1. What kind of invisible light does a **PIR** sensor feel?
2. Why does it notice a **moving** person but not a still one?
3. Where have you seen motion sensors?

Scan to watch

After the video: Plan the best spot in your room for a burglar alarm sensor. Draw it!

The science

Your body glows (in infrared)!



Invisible glow

Every warm thing gives off **infrared** light: people, cats, a hot cup of tea. Our eyes cannot see it, but the PIR sensor can.

It spots CHANGES

The PIR does not see a still picture. It notices when warm infrared **moves** across its view. Stand still and it may stop seeing you!

Remember JM-06?
The flame sensor also saw infrared. Same light, different job!

PIR means
Passive InfraRed. Passive = it only looks, it sends nothing out.

Real world

Sensors and secret codes everywhere



Home alarms
Motion sensors in corners, a keypad by the door



Security lights
Switch on outside when someone walks past at night



Phone lock
A PIN or pattern: the order of digits matters



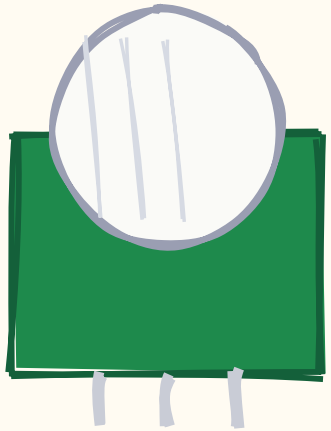
Bank cards
A 4-digit PIN at the ATM keeps your money safe

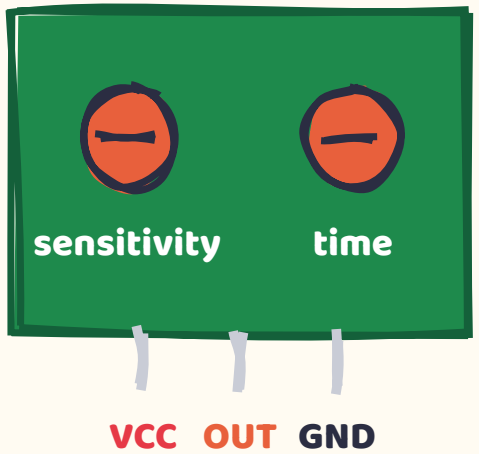
? Look around the school. Can you find a **motion sensor**? (Hint: corridor lights, the main gate, automatic doors.)

Two parts
SENSE someone is there + CHECK who it is. Our alarm does both!

Meet the sensor

The PIR motion sensor





The two orange dials

Sensitivity: how far away it can see.
Time: how long OUT stays HIGH after movement. Turn it fully **anticlockwise** for the shortest time.

The white dome is a special lens that collects infrared from a wide area.

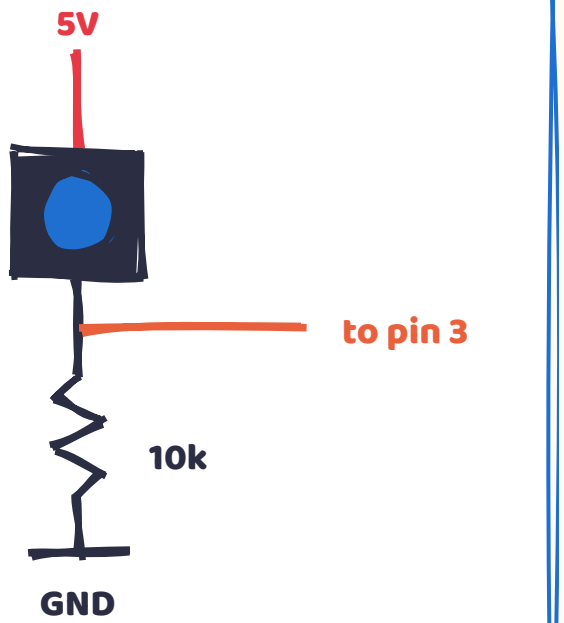
Pin	What it does
VCC	5V
OUT	HIGH when it sees movement
GND	Ground

Be patient!

After power on, the PIR needs about **1 minute** to settle. It may give false alarms until then.

Meet the input

Buttons need a helper resistor



The floating problem

When a button is **not pressed**, the pin is connected to nothing. It "floats" and may read HIGH or LOW at random, like a ghost pressing it!

We need 3

Button 1 → pin 3
Button 2 → pin 4
Button 3 → pin 5
Each with its own 10k to GND.

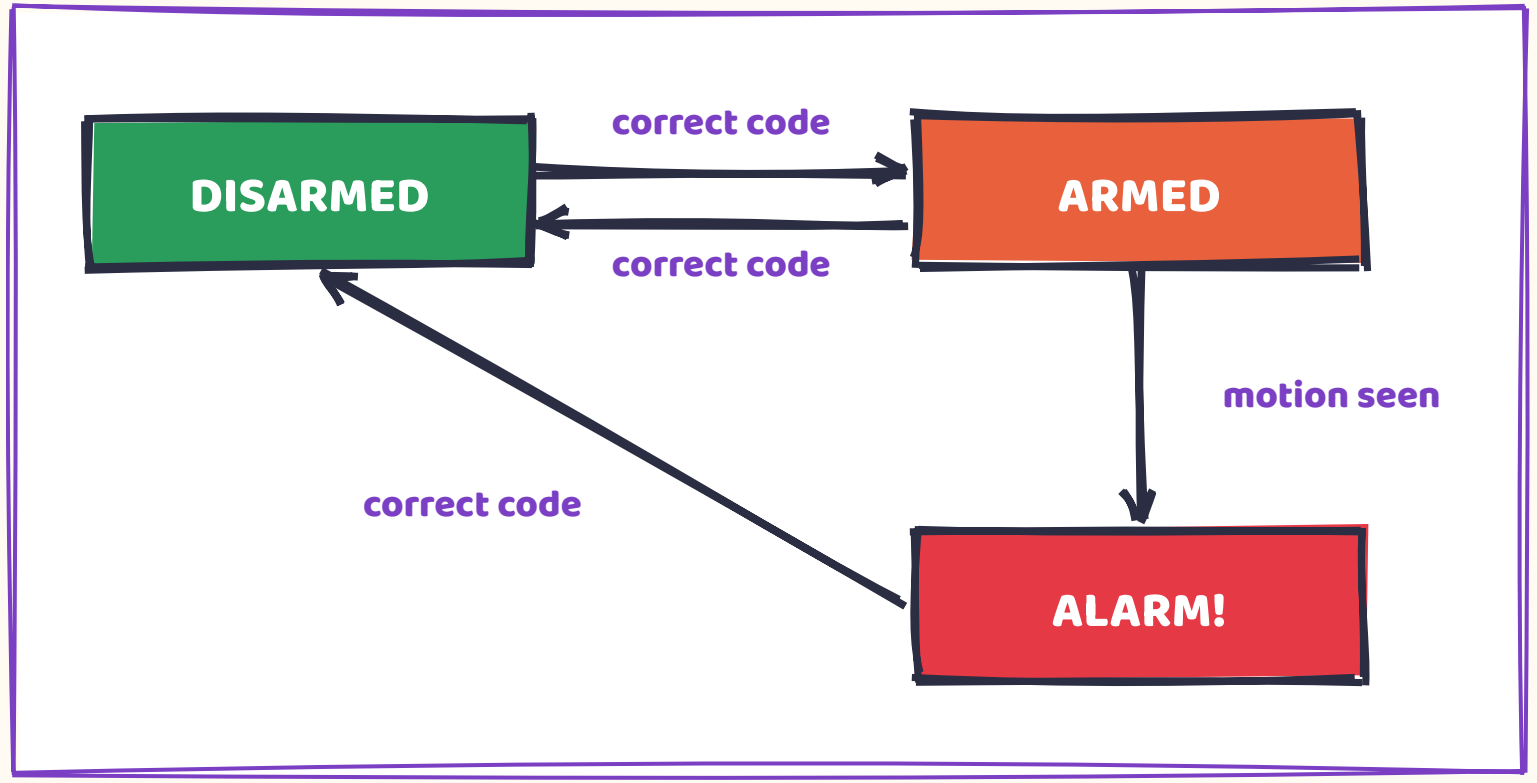
The fix: pull-down

A **10k resistor** gently pulls the pin to GND, so it reads **LOW**. Press the button and 5V wins: **HIGH**.

Tip
Put buttons ACROSS the middle gap of the breadboard, so the legs are not joined.

Big idea

An alarm has STATES



What each state does

DISARMED: green LED, ignore motion

ARMED: red LED, watching

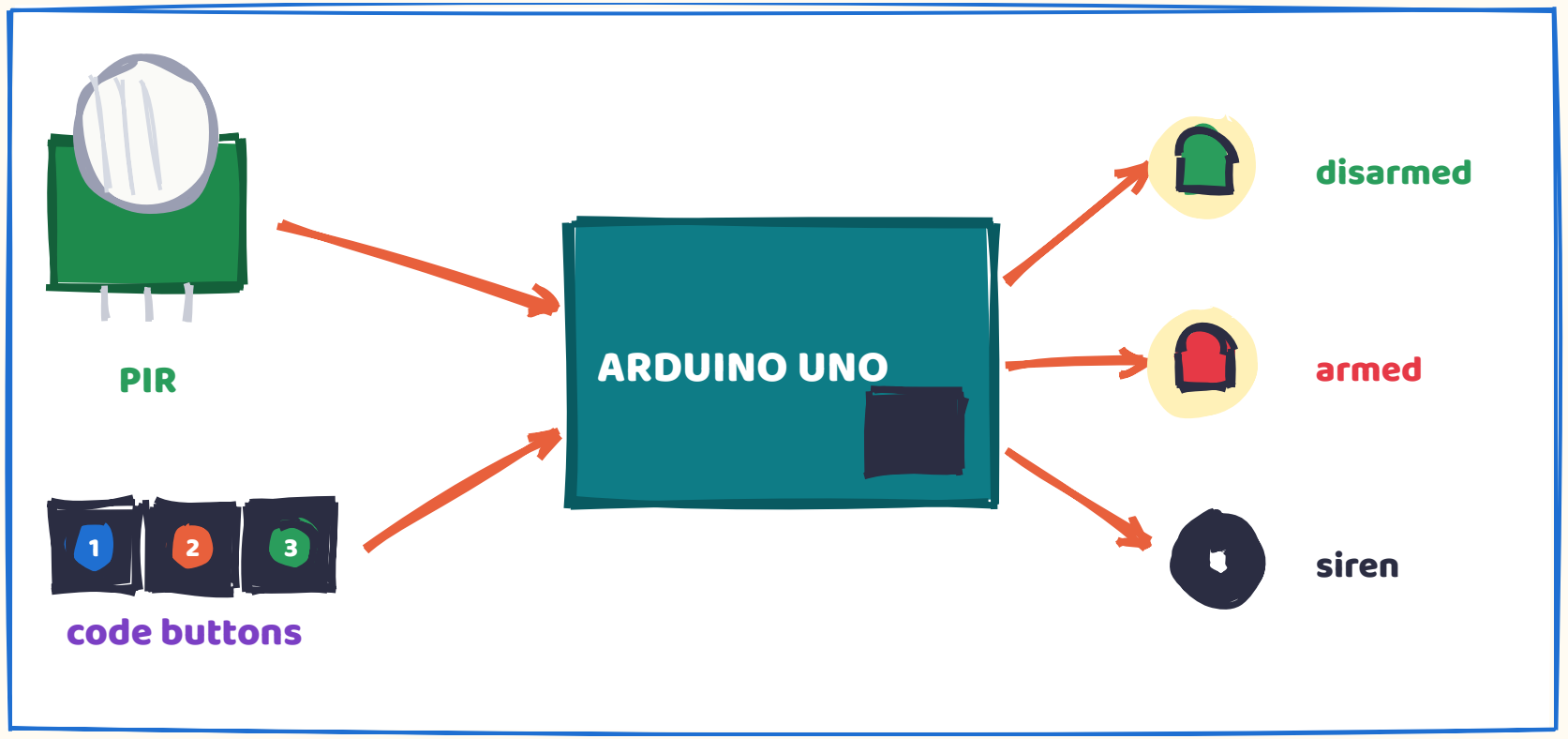
ALARM: red LED + siren until the code is entered

State variable

We store the state in a variable called armed: 0 = no, 1 = yes.

The plan

Sense, check, warn



Inputs and outputs

Inputs: 1 PIR + 3 buttons

Outputs: 2 LEDs + 1 buzzer

The **brain** decides using the state and the code.

Hide the buttons!

Put the keypad near the door, but where a burglar would not see you press it.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. How many 3-button codes are there?
2. Crack the code!
3. Our alarm kit (per group)
4. Wiring the burglar alarm
5. The alarm in ArduinoBlocks
6. The secret code checker
7. Is our alarm secure?

Maths moment

How many 3-button codes are there?

Order matters!

1-2-3 and **3-2-1** use the same buttons but are **different codes**. That is called a **sequence**.

`111` `123` `321` `231` `...` `333`

Challenge

With **4** presses: 3×3×3×3 = ?

A phone PIN has **10** digits and **4** presses: 10×10×10×10 = ?

1st press: **3** choices × 2nd: **3** × 3rd: **3** = **27 codes**

Answers

81 codes. 10,000 PINs! Longer codes are much harder to guess.

Pair game

Crack the code!

How to play

1. Partner A secretly writes a 3-button code, like **2-1-3**.
2. Partner B "presses" buttons by saying a number.
3. After each press A says only **"yes"** or **"no, start again"**.
4. Count how many presses B needs to crack it.
5. Swap and try a 4-button code!

Then discuss

Saying "yes" after each press made it **easy** to crack. Should a real alarm tell a burglar which presses were right?



Big idea

Good security gives away as little information as possible.

Build it

Our alarm kit (per group)

Arduino Uno
The brain

PIR sensor
Spots movement

3 × push button
With 3 × 10k resistors

Green + red LED
With 2 × 220 Ω

Buzzer
The siren

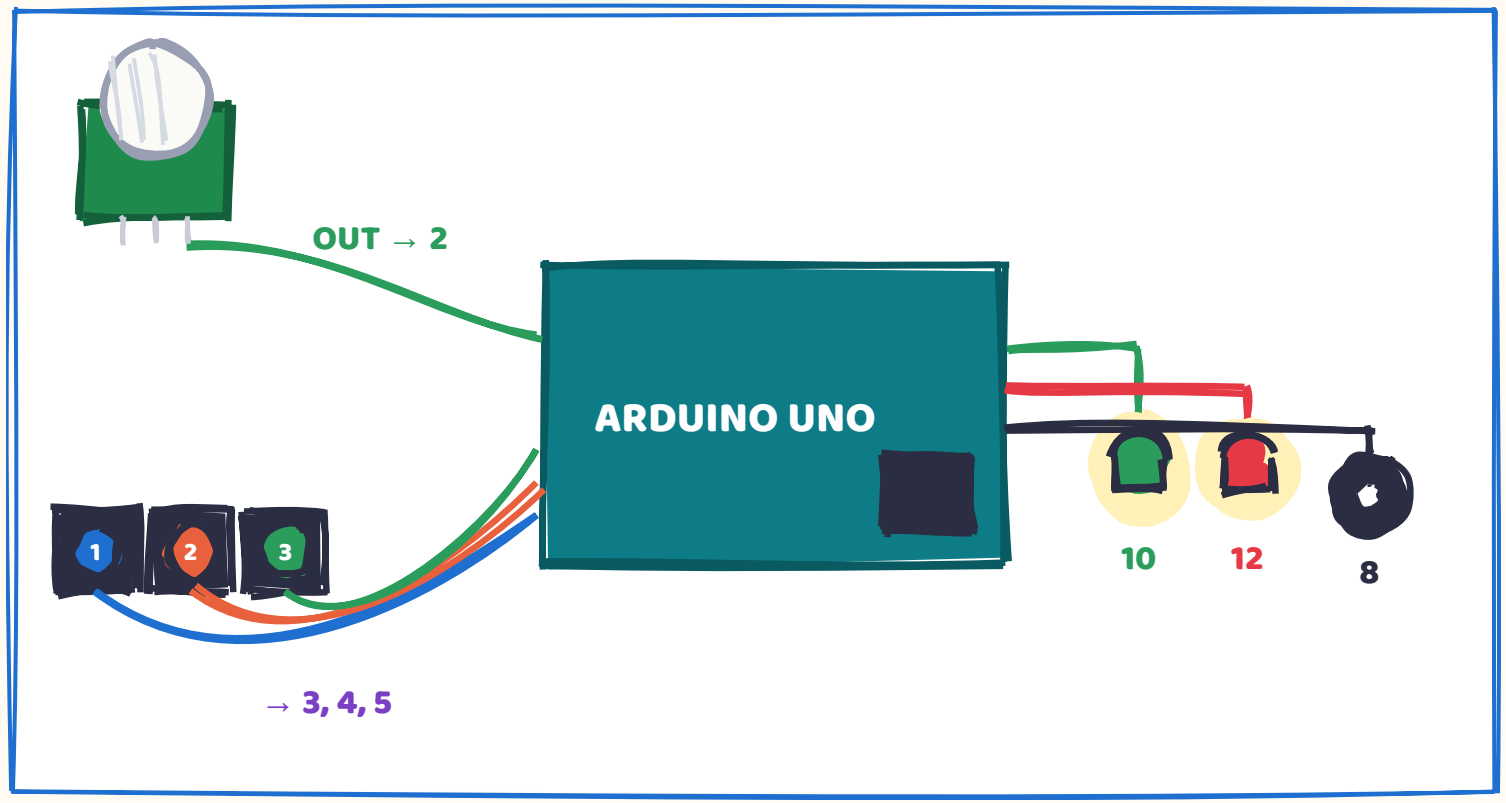
Card box
A model room or house

Also: breadboard, about 14 jumper wires (M-M), 3 M-F wires for the PIR, USB cable, laptop.

Re-use
LEDs and buzzer work just like the safety panel in JM-06.

Build it

Wiring the burglar alarm



Part	Arduino
PIR VCC / OUT / GND	5V / pin 2 / GND
Button 1, 2, 3	pins 3, 4, 5
Other button legs	5V
3 × 10k resistors	pin to GND
Green LED (+ 220 Ω)	pin 10
Red LED (+ 220 Ω)	pin 12
Buzzer +	pin 8

Check twice
A button wired the wrong way round is always ON. Test each one alone first!

Code it: part 1

The alarm in ArduinoBlocks

```
when Arduino starts
    armed = 0 · alarm = 0 · step = 0 · wait 60 s (PIR settles)
forever
    check the secret code (next slide)
    if armed = 1 and read digital pin 2 = HIGH
        set alarm to 1 (stays on!)
    if alarm = 1
        red ON · tone pin 8, 1500 Hz, 150 ms · wait 150 ms
    else if armed = 1
        red ON · green OFF · no tone
    else
        green ON · red OFF · no tone
```

Read it like a story

Check the code buttons.
Armed + motion? Set **alarm** to 1.
Alarm stays on even if the burglar **stands still**, until the code is entered.
Not armed? Green, quiet.



Test it fast
While testing, change 60 s to 10 s. Put it back for the real thing!

Code it: part 2

The secret code checker

```
our code is 2 → 1 → 3
if button 2 pressed and step = 0
    set step to 1
else if button 1 pressed and step = 1
    set step to 2
else if button 3 pressed and step = 2
    set armed to 1 - armed (flip it!) · set alarm to 0
    set step to 0 · beep beep
else if ANY button pressed
    set step to 0 (start again)
if any button was pressed: wait 300 ms
```

The step counter

step remembers how many correct presses so far.
0 = nothing yet · 1 = first one right · 2 = two right.
One wrong press? Back to **0**.

Flip it!

1 - armed turns 0 into 1 and 1 into 0. The same code arms **and** disarms, and always silences the siren.

Why wait 300 ms?
One press can look like many to a fast Arduino. The wait stops double counting.

Test like an engineer

Is our alarm secure?

Test	Expect	Result
Disarmed, wave hand at PIR	green, quiet	pass / fail
Press 2-1-3	beep beep, red ON	pass / fail
Armed, wave hand	SIREN!	pass / fail
Press 2-1-3 during siren	green, quiet	pass / fail
Press 1-2-3 (wrong)	nothing changes	pass / fail
Press 2-1-1-3	nothing (step reset)	pass / fail

Debug checklist

Siren never stops? Turn the PIR time dial anticlockwise.
Random alarms? Wait the full minute.
Code never works? Print step to the Serial monitor.

Swap with another team. Can they get in **without** knowing your code?

Be a good digital citizen

Keep your codes secret

Don't use easy codes: 1-1-1, 1-2-3, your birthday.

Don't share your code or password, even with a best friend.

Watch out for people looking over your shoulder (shoulder surfing).

Do use longer codes and passwords that are hard to guess.

Do tell a trusted grown-up if you think someone knows your password.



Same rules

for your alarm code, phone PIN, school login and games account.

Level up!

Alarm challenges

Choose your challenge



Entry delay: when motion is seen, flash red for 10 seconds before the siren, so family has time to type the code.



Lockout: after 3 wrong codes, ignore all buttons for 30 seconds.



Longer code: make it a 4-press code. How many codes now?



Feedback: a short click for every press, so the user knows it was counted.

Think like an engineer

Our alarm can be beaten by **pulling out the USB cable**! How do real alarms stop that?

Boss level

Answer: a backup battery inside, and a siren that sounds if the box is opened.

Show what you know

What did we learn?

- [x] A PIR sees moving infrared from warm bodies
- [x] Pull-down resistors stop buttons from floating
- [x] An alarm has states: disarmed, armed, alarm
- [x] A step counter checks a code where order matters
- [x] Long, secret codes are much harder to crack

Exit ticket: how many different 3-press codes can we make with 3 buttons?

At home

With a grown-up, count the motion sensors and keypads you see in one trip to a shop or bank.



Next class: 3D-Designed Name Badge & Sensor Mount!